SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

Code of Conduct:

I **M Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

Comparative Analysis

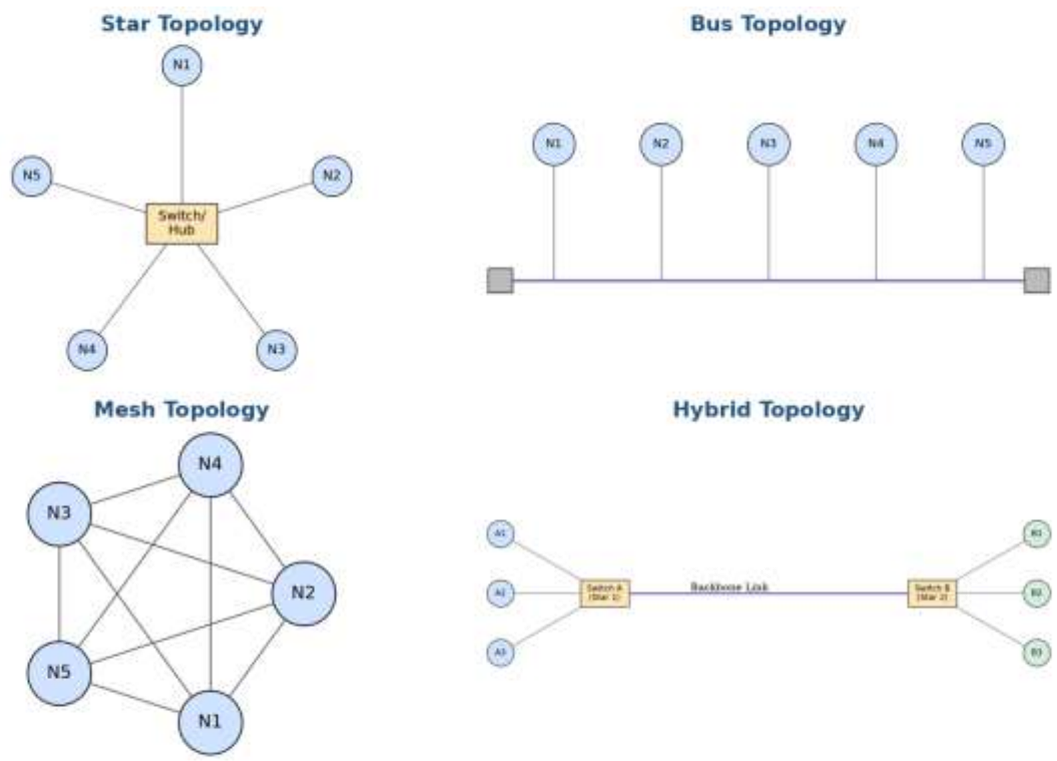
1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are

difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – CSA0706 COMPUTER NETWORKS (CO4)

Reference: The Four Topologies Compared in This Report



1. Star vs Bus Topology – Cost, Reliability, and Performance

Given (Original Statement): Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Criterion	Star Topology	Bus Topology
Cost	Higher – requires a central device (switch/hub) plus an individual cable run to every node.	Lower – uses a single shared backbone cable with minimal connection hardware; cheapest to install.
Reliability	Higher for node failures – one node/cable failing does not affect any other node. Central device is a single point of failure.	Lower – a break anywhere on the shared backbone cable brings down the entire network.
Performance	Better – the central device manages traffic and reduces collisions between nodes.	Degrades as more devices are added – all nodes share one medium, so traffic and collisions increase with network size.

Conclusion: Star topology trades higher upfront cost for significantly better reliability and performance, making it the stronger overall choice once a network grows beyond a handful of nodes – bus topology's low cost only remains attractive for small, non-critical, budget-constrained networks.

2. Mesh vs Star Topology – Fault Tolerance

Given (Original Statement): Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Criterion	Mesh Topology	Star Topology
Path Redundancy	Every node connects to multiple other nodes, so multiple alternate paths exist for any given transmission.	Every node connects only to the central hub/switch – there is exactly one path between any two nodes.
Link Failure	If one link fails, data is simply routed through an alternate path with no disruption to communication.	If an individual node's link fails, only that node is affected; other nodes continue normally.
Central-Point Failure	No single central device exists, so there is no single point of failure at all.	Failure of the central hub/switch brings down the entire network – every node loses connectivity simultaneously.
Fault Tolerance Rating	Very high	Moderate

Conclusion: Mesh topology is more robust and reliable for critical environments (e.g., financial trading floors, military or emergency communication networks) precisely because it has no single point of failure, whereas star topology's dependence on one central device makes it a weaker choice wherever continuous availability is essential.

3. Bus vs Mesh Topology – Installation Complexity

Given (Original Statement): Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Criterion	**Bus Topology**	**Mesh Topology**
Cabling Required	Minimal – a single backbone cable with a short drop cable per node.	Very high – every node must be individually cabled to every other node.
Connections per Node (n nodes)	1 connection per node to the shared backbone (n connections total).	$(n - 1)$ connections per node – $n(n - 1)/2$ links overall for full mesh.
Growth Behavior	Adding a node is simple – one new drop cable to the backbone.	Adding a node requires a new cable/port to every existing node – complexity grows quadratically.
Best Suited For	Small networks where simplicity and low install time matter most.	Small numbers of critical nodes where redundancy justifies the wiring cost/effort.

Conclusion: Bus topology is dramatically simpler and faster to install than mesh, and that gap widens quickly as the network grows – mesh's cabling need grows quadratically ($n(n-1)/2$ links) while bus grows only linearly (n links), making mesh time-consuming and costly to scale despite its reliability advantage.

4. Hybrid vs Star Topology – Scalability

Given (Original Statement): Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Criterion	Hybrid Topology	Star Topology
Structure	Combines two or more topologies (e.g., star + mesh, star + bus) into one flexible network.	A single central device connects all nodes directly.
Expansion Method	New segments/topologies can be added without disturbing the existing structure (see Figure 0.1's Switch A / Switch B hybrid example).	New nodes are added directly to the same central device.
Limiting Factor	Limited mainly by how many segments can be interconnected – highly flexible.	Limited directly by the port capacity of the central hub/switch.
When Capacity Is Reached	A new segment (with its own switch/hub) is simply added and linked in.	Expansion requires purchasing and installing additional central hardware, and possibly redesigning the wiring.
Scalability Rating	High	Limited / Moderate

Conclusion: Hybrid topology offers substantially greater scalability than star topology because it grows by adding whole new segments rather than being bounded by a single device's port count – making it the better long-term choice for organizations that expect continued growth.

5. Maintenance Complexity Across All Four Topologies

Given (Original Statement): In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Topology	Maintenance Characteristics	Difficulty
Star	Issues can be quickly identified and isolated to a specific node or cable, since each connects independently to the central device.	Easy
Bus	A fault in the shared backbone cable can disrupt the entire network and is difficult to physically locate along the cable run.	Hard
Mesh	Highly reliable once running, but its large number of interconnections makes troubleshooting which specific link has failed time-consuming.	Hard
Hybrid	Benefits from the flexibility of combined topologies, but managing multiple combined structures requires careful monitoring and skilled administration.	Moderate

Conclusion: Ranked from easiest to hardest to maintain: Star (easiest, faults isolate cleanly) < Hybrid (moderate, needs skilled oversight of combined segments) < Mesh ≈ Bus (hardest, though for different reasons – mesh from sheer connection volume, bus from difficulty locating a single backbone fault).

Overall Comparative Summary

Bringing all five comparisons together, each topology has a distinct profile rather than a single "best" answer – the right choice depends on which criterion matters most for a given deployment:

Topology	Cost	Reliability / Fault Tolerance	Performance	Install Complexity	Scalability	Maintenance
Star	Higher	Moderate	Good	Moderate	Limited	Easy
Bus	Lowest	Low	Degrades with growth	Simplest	Limited	Hard
Mesh	Highest	Very High	Good (multiple paths)	Very Complex	Poor (quadratic cabling)	Hard
Hybrid	Variable	High (inherits best of parts)	Good	Moderate–Complex	High	Moderate

In practice: bus suits small, low-budget, non-critical networks; star suits typical office/departmental LANs needing easy management; mesh suits small numbers of mission-critical nodes (e.g., core routers, data-center backbones) where redundancy justifies the cost; and hybrid suits growing organizations that need the manageability of star segments combined with the flexibility to expand or add redundancy where it matters most – which is why hybrid designs are the most common choice in real-world enterprise networks.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion